Workshop #1: Welcome to the R environment

What is R and how

do we open it, why would

we use it, just why

Presentation contents

What is R/ R studio

- Objects
- Functions
- Scripts
- Packages
- Paths
- Comments

Learning objectives

- Be able to save data on your computer and read data into R studio
- Coping with error messages
- Understand the definitions of:
    - Objects
    - Functions
    - Scripts
    - Packages
    - File paths
    - Comments

Long term workshops

- Session 1. (May 2026): R basics, including what each window pane means, different menus, syntax, and packages
- Session 2. (August 2026): R basics review, introduction to data cleaning and variable creation
- Session 3. (August 2026): Intermediate data cleaning, joining datasets, and basic statistical analysis in R
- Session 4. (September 2026): Data visualization with ggplot2
- Session 5. (September 2026): Automating reports with R markdown and integration with GitHub
    - Survey weights here as well?

Class Repository

- https://github.com/SamSwift505/NMDOH_R_WORKSHOPS_26
- All materials will be available here

Installing R and R studio

- We need R and R studio to do this!
- Instructions on how to install both R and R studio are below points A1-A3 here:
- [https://rstudio-education.github.io/hopr/starting.htmlLinks to an external site.](https://rstudio-education.github.io/hopr/starting.html)
- A video on installing R and R studio
- https://youtu.be/Tb3R4GLJ45U

To run R Studio on your computer, you need to have both R and R Studio installed.

- Both of these are applications that you download
- You need admin privileges!

Why do we need statistical software?

- Excel is nice when you have <30 values or need to do record keeping tasks
- NOT NICE for larger scale data manipulation or analysis
- Statistical software provides much more control
- Statistical software ensures reproducible science
  - I often submit my code when I publish


This article is more than **4 years old**

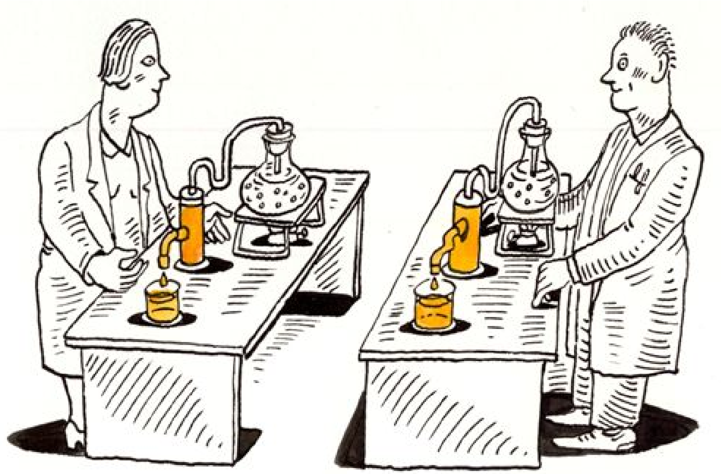
Analysis

Covid: how Excel may have caused loss of 16,000 test results in England

Alex Hern

UK technology editor

Public Health England data error blamed on limitations of Microsoft spreadsheet



Entertaining meme

What is R?

Language & Environment

Statistics

FREE!

Community

Customizable

Online Training

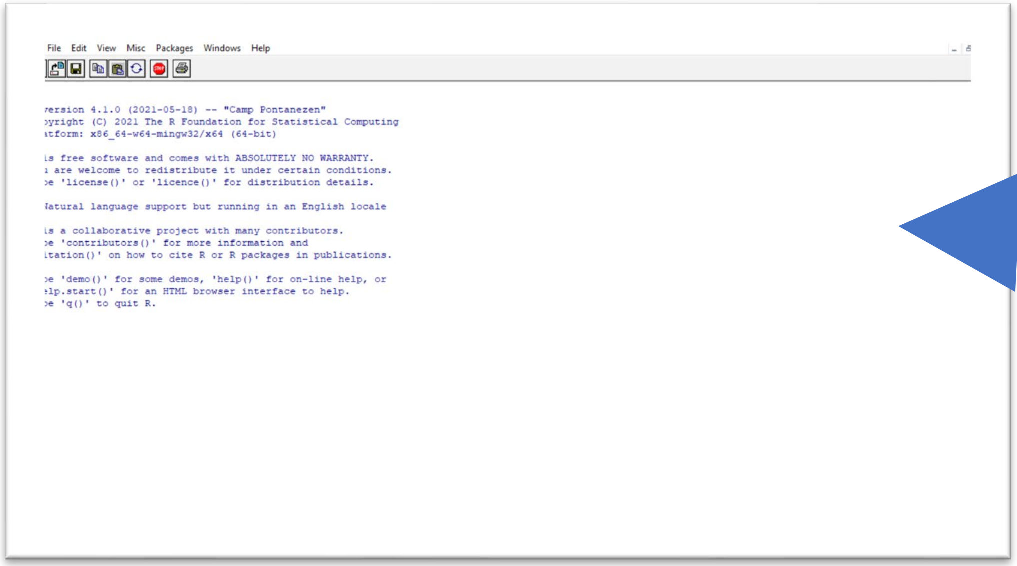
https://www.r-project.org/

There are Some Disadvantages to R/ R studio

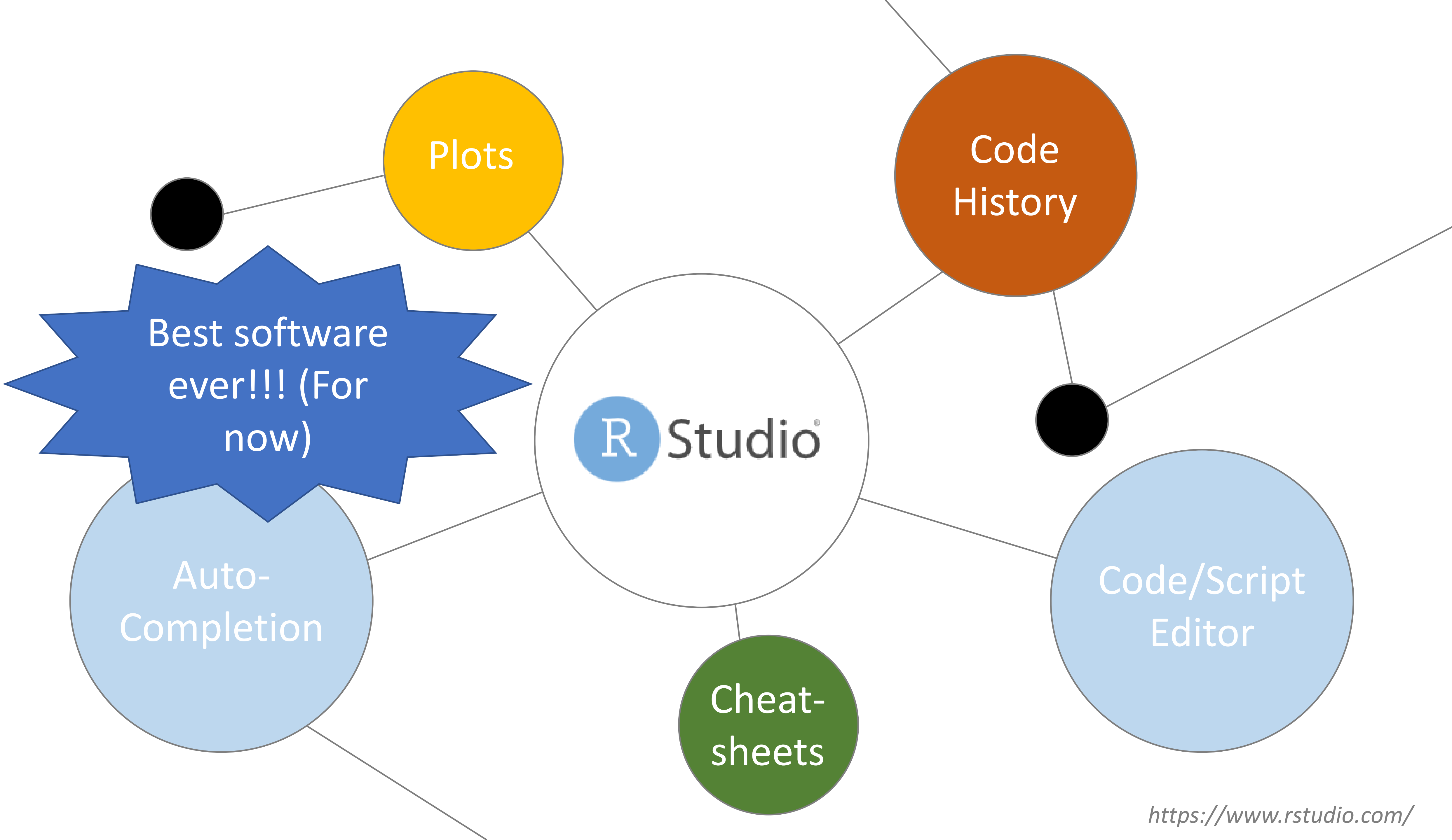
- Can be less intuitive to learn then other programs
- Updates require maintenance
- Security
- Can be slower than other languages
- Some packages are not great quality
- Memory management

R vs R studio- Why do we need both?

- By installing R, you install the code that R studio runs on on your computer
- R studio us a “user friendly” (?) interface that we use to code in R



It is possible to work directly in the R Console, but it is clunky!



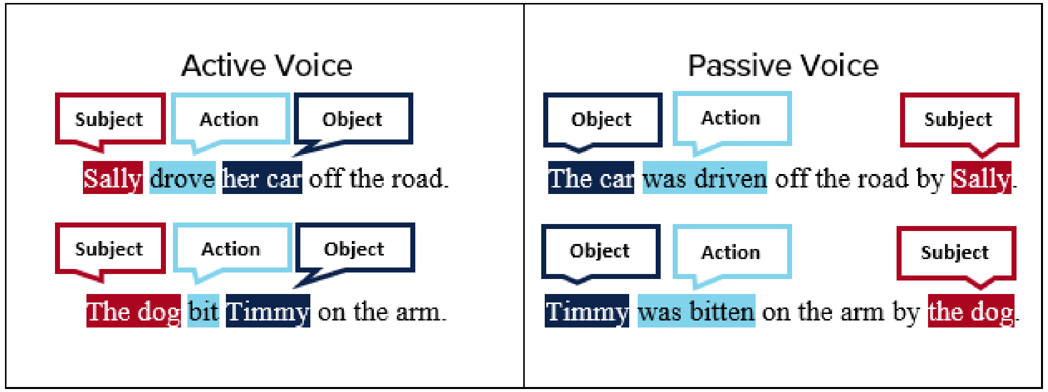
https://www.rstudio.com/

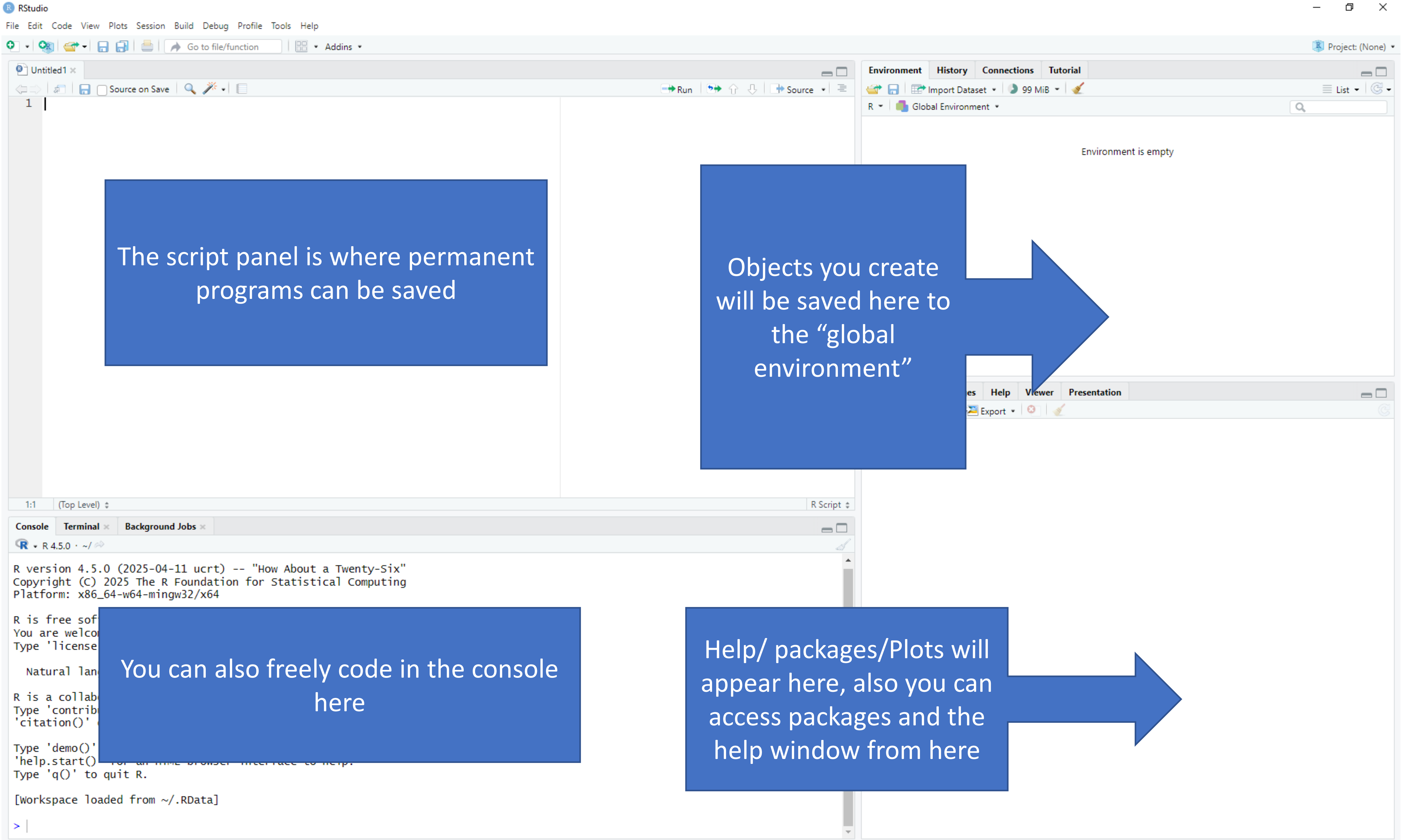
Code has syntax

- Think of lines of code like learning a new language
- To teach you this I do a special segment called

Syntax Breakdown

- By the end of this class you will know some basic words and structure, I do not expect you to speak the language





RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Untitled1
Source on Save
Run
Source
Environment History Connections Tutorial
Import Dataset
99 MiB
List
Global Environment
Environment is empty
The script panel is where permanent programs can be saved
Objects you create will be saved here to the "global environment"
Help Viewer Presentation
Export
1:1 (Top Level)
R Script
Console Terminal Background Jobs
R 4.5.0 · ~/
R version 4.5.0 (2025-04-11 ucrt) -- "How About a Twenty-Six"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64
You can also freely code in the console here
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
Help/ packages/Plots will appear here, also you can access packages and the help window from here

Key Terms

- **Objects** - Everything you store in R - datasets, variables, a list of village names, a total population number, even outputs such as graphs - are objects which are assigned a name and can be referenced in later commands.

- **Functions** - A function is a code operation that accept inputs and returns a transformed output.

- **Packages** - An R package is a shareable bundle of functions.

- **Scripts** - A script is the document file that hold your commands.

Objects

- Objects are things you make, like datasets, tables, graphs and more
- The arrow operator (less than, dash) is what we use to assign things into objects
- That operator looks like this

<-

The syntax we will use to make objects will look like this:

object_name <- **value**

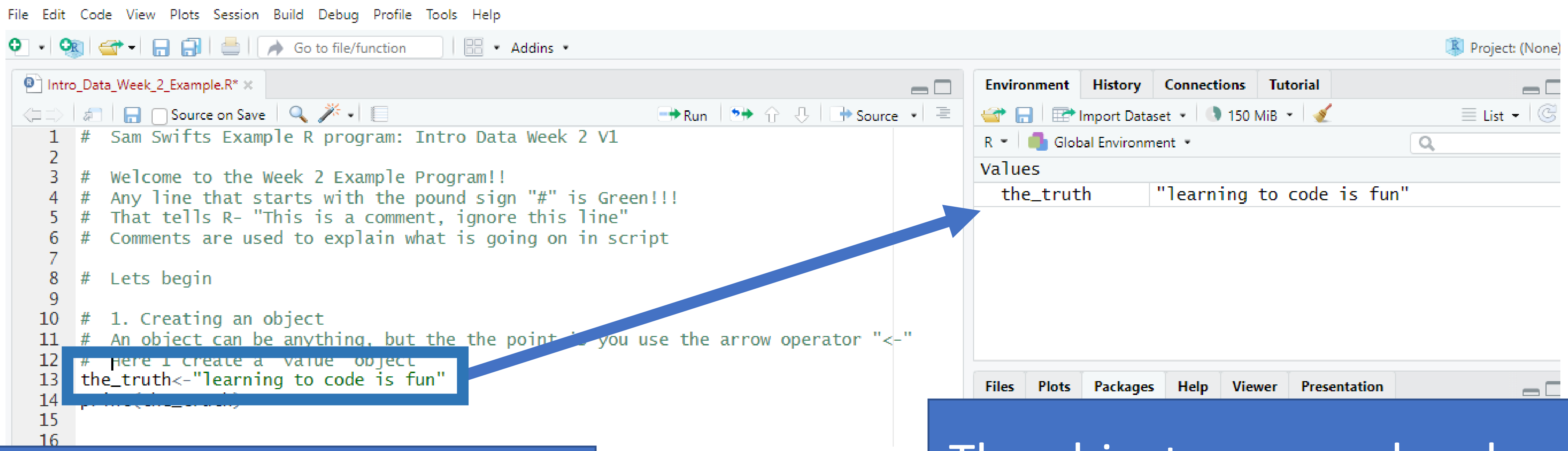
Objects Continued

- There are lots of different types of objects
  - We will mostly use data frames (datasets)
  - And plots (graph/plots)
  - Today I will show a simple "value" object
- There are also many other more advanced ones
  - Vectors
  - Matrices
  - Lists (data frames are special lists)
  - Arrays

Don't worry about memorizing these, just know that in R we make objects

Objects Continued

- When you make an object, it will appear in the top right “environment window”



I ran this code

The object appeared up here in “Environment”

Syntax Breakdown

Break it down

```
the_truth<-"learning to code is fun"
```

Like parenthesis, if you open quotes you must close them

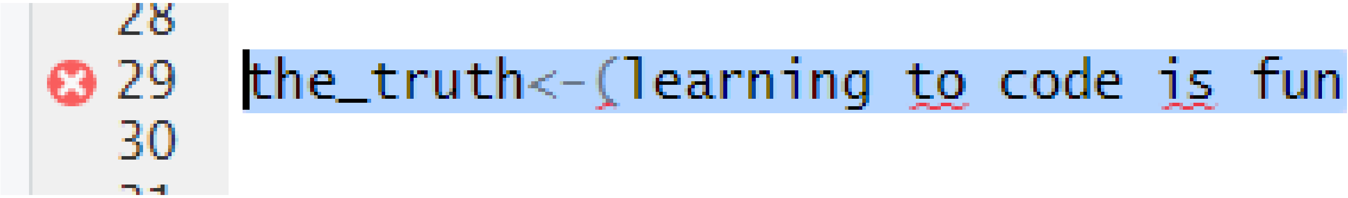
The object we created, can be named anything AS LONG AS THERE ARE NO SPACES

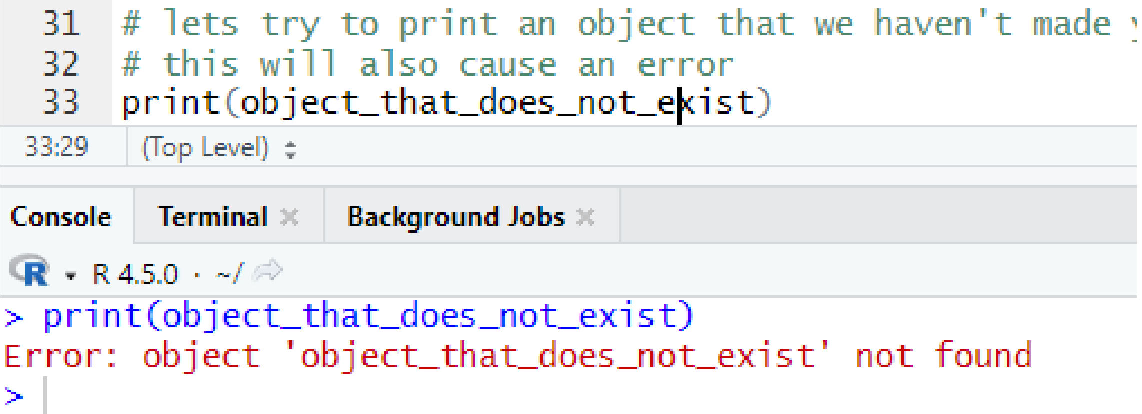
A string of text- the computer knows its text because its in quotes

Our friend the arrow operator, says take what follows and put it in the object

Error messages

- If you make a mistake, R will error out!!!
- Usually this happens with angry red text in the console window, or an X on the line where the error is





Errors are your friends– they help you learn!!

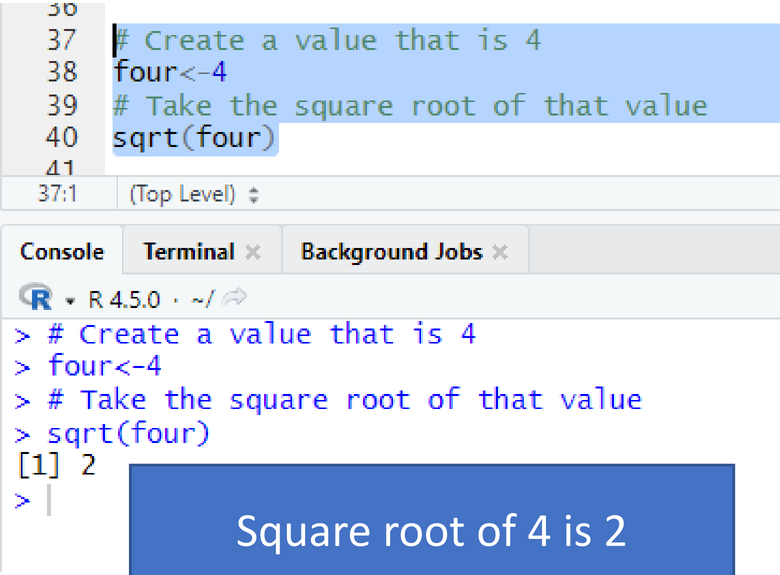
Debugging Errors

- Our class is a community of practice
- Lets work together in class discussions to help
- Do not be embarrassed
- If no one helps I will help
- You can also paste errors into google (which I will demonstrate)
- You can also ask AI but its often wrong

Functions

- Functions preform tasks and operations
- A function often wraps around an object, using parenthesis
- Example:

Break it down



Square root of 4 is 2

sqrt(four)

An object I made that is the number 4

The function– Note the parenthesis wrapping the object

Packages

- From your reading:
- "An R package is a shareable bundle of code and documentation that contains pre-defined functions."
- This is the most exciting thing about R!!
- In this class we will use three:
  - tidyverse (next week)
  - dplyr (next week)
  - ggplot2

Ok… so what is a package??

- Basically installing a package is like downloading more functions within R studio
- Once you install ggplot2, you get access to a lot of amazing data visualization functions
- Once you download it, its there forever, but you need to ask R studio to load it in each session (shown in 2 slides)

State adoption of paid sick leave and cardiovascular disease…

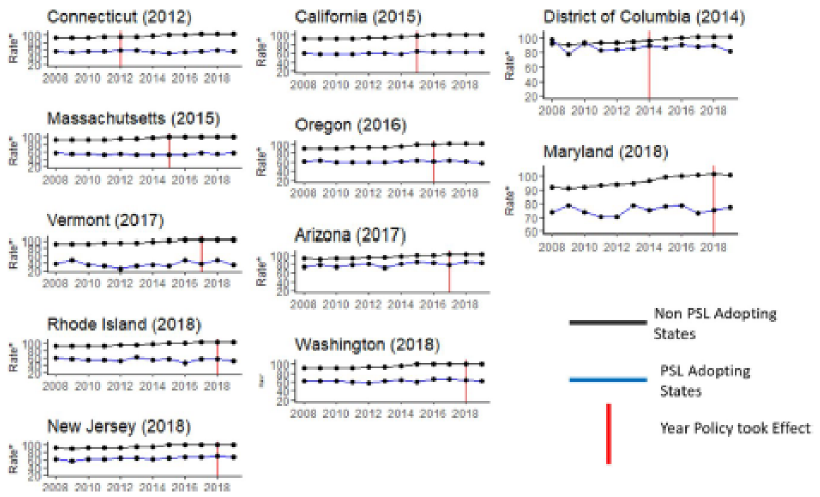
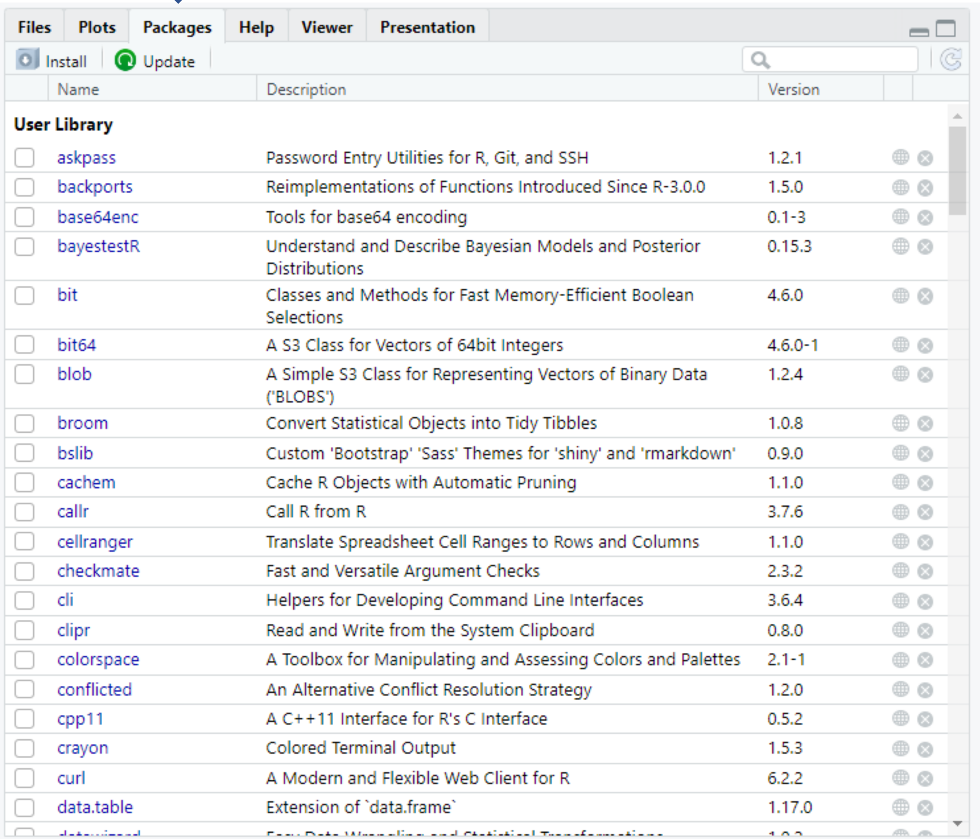


Fig. 2 Annual cardiovascular disease mortality rate among persons 15–64 in states that adopted State-level paid sick leave policies (treated) vs non-adopting (NA) states (untreated), 2008 to 2019. Rates are county-level mortality rates due to cardiovascular disease among persons 15–64, per 100,000 persons. Non-adopting states (as of 2019) in the Northeast are Maine, New Hampshire, Pennsylvania, and New York. Non-adopting states (as of 2019) in West are Colorado, Idaho, New Mexico, Montana, Utah, Nevada, Wyoming, Alaska, and Hawaii. Non-adopting states (as of 2019) in the South are Delaware, Florida, Georgia, North Carolina, South Carolina, Virginia, West Virginia, Alabama, Kentucky, Mississippi, Tennessee, Arkansas, Louisiana, Oklahoma, and Texas

Downloading Packages

In the "packages" tab you can search for and download packages



Files | Plots | Packages | Help | Viewer | Presentation

Install | Update

Name	Description	Version
User Library		
askpass	Password Entry Utilities for R, Git, and SSH	1.2.1
backports	Reimplementations of Functions Introduced Since R-3.0.0	1.5.0
base64enc	Tools for base64 encoding	0.1-3
bayestestR	Understand and Describe Bayesian Models and Posterior Distributions	0.15.3
bit	Classes and Methods for Fast Memory-Efficient Boolean Selections	4.6.0
bit64	A S3 Class for Vectors of 64bit Integers	4.6.0-1
blob	A Simple S3 Class for Representing Vectors of Binary Data ('BLOBS')	1.2.4
broom	Convert Statistical Objects into Tidy Tibbles	1.0.8
bslib	Custom 'Bootstrap' 'Sass' Themes for 'shiny' and 'rmarkdown'	0.9.0
cachem	Cache R Objects with Automatic Pruning	1.1.0
callr	Call R from R	3.7.6
cellranger	Translate Spreadsheet Cell Ranges to Rows and Columns	1.1.0
checkmate	Fast and Versatile Argument Checks	2.3.2
cli	Helpers for Developing Command Line Interfaces	3.6.4
clipr	Read and Write from the System Clipboard	0.8.0
colorspace	A Toolbox for Manipulating and Assessing Colors and Palettes	2.1-1
conflicted	An Alternative Conflict Resolution Strategy	1.2.0
cpp11	A C++11 Interface for R's C Interface	0.5.2
crayon	Colored Terminal Output	1.5.3
curl	A Modern and Flexible Web Client for R	6.2.2
data.table	Extension of `data.frame`	1.17.0

You install it once using install.packages function

- You can also type:

```
install.packages("<package name>")
```

into the console

- Once the package is downloaded, the contents can be made available by using:

```
library(<package name>)
```

You call the package in using the library function in every time

SYNTAX BREAKDOWNNNN

Function to install the first time

```
install.packages("tidyverse")
# here is the library function
library(tidyverse)
```

This is a comment, a line that does nothing but is used for documentation/ notes (starts with a hashtag)

Function to read package in every time- you must do this at the beginning of every R session

The package we are installing/ using

Commonly Used Packages

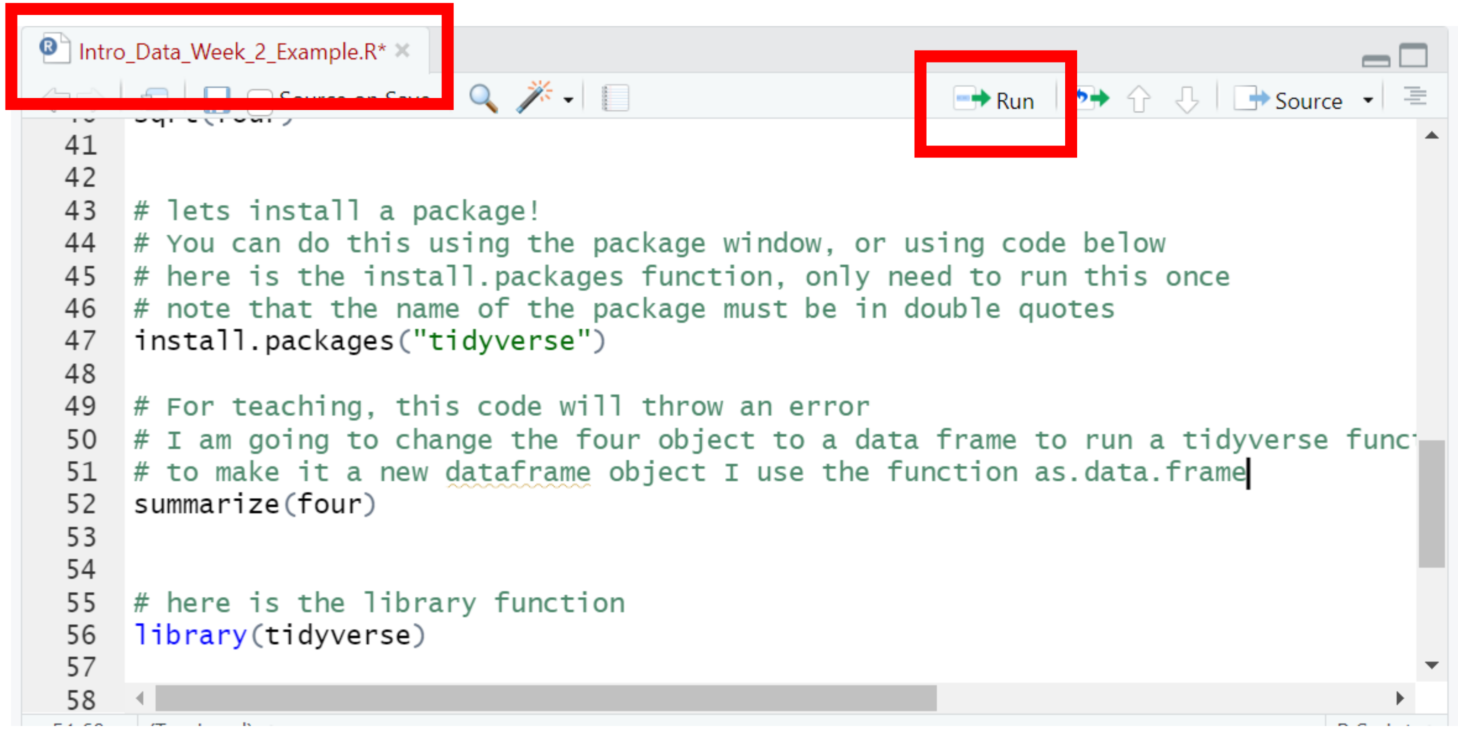
- **tidyverse**: A collection of packages designed for data manipulation
- **dplyr:** Shortcuts for sub-setting, summarizing, rearranging, and joining data
- **tidyr:** Changing the layout of the data sets
- **stringr:** Tools for regular expressions and character strings
- **lubridate**: Makes working with dates and times easier
- **ggplot2:** Create graphics
- **R Markdown:** Can be used for automated reporting. Reports can be exported as HTML, pdf, or MS Word document

Scripts

- By coding in the "scripts" window you have a file that you can come back to in later sessions
  - All statistical software has something like this
  - You will be required to save an upload a script for the homework

Red text means your version is unsaved

The run button will run highlighted code



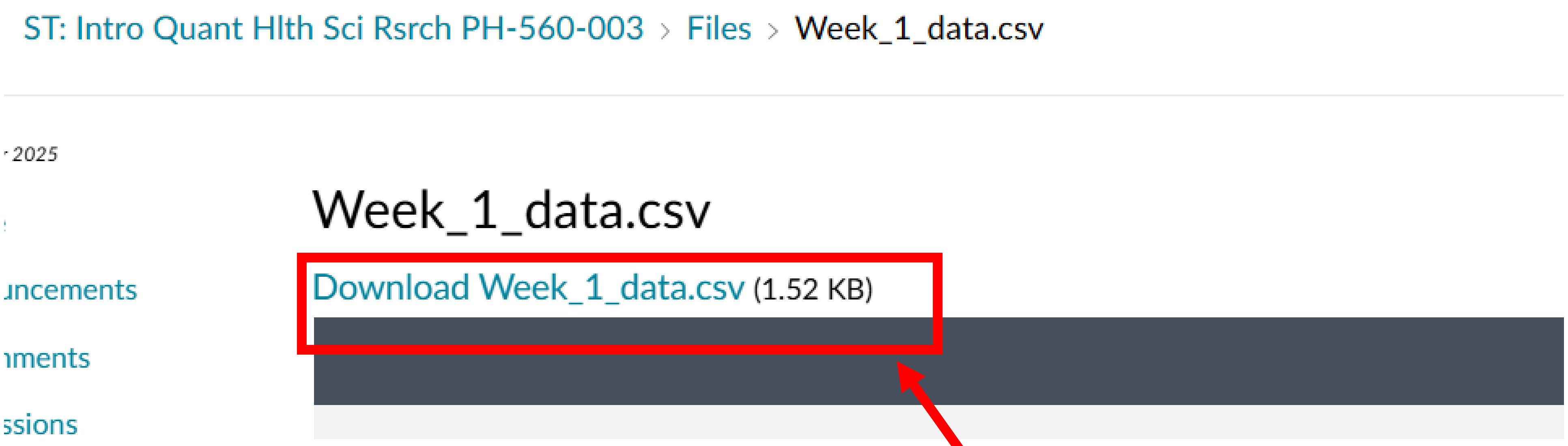
File paths and reading in data

- One of the hardest things for new users of statistical software is reading in data
- For this class, there are three broad steps

1. Download the data from Canvas (demonstrated in video today)
2. Save the data in an organized place on your computer
3. Use code to tell R studio where the data is and read it in as a new object.

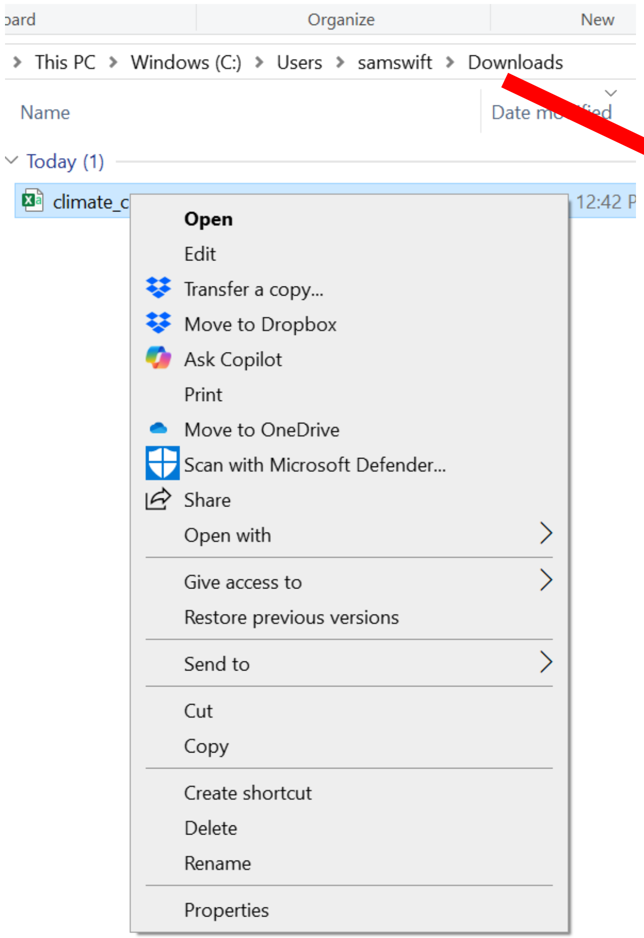
File paths are a key concept for the week- we use these to read data in and out

Downloading the data from Canvas

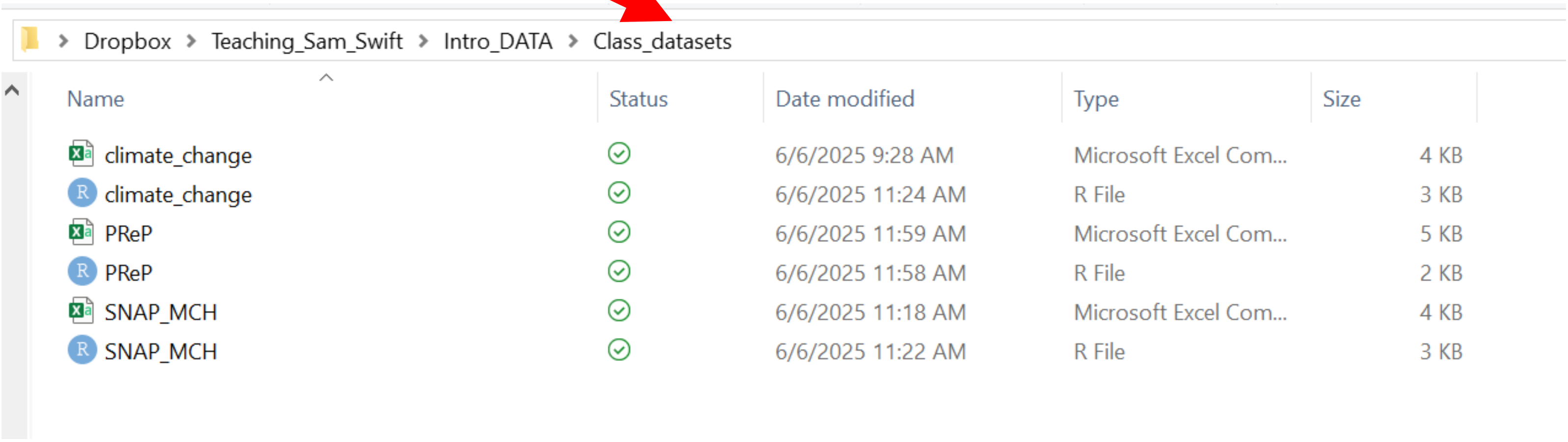


Double clicking this link downloads your data

Save the file somewhere organized

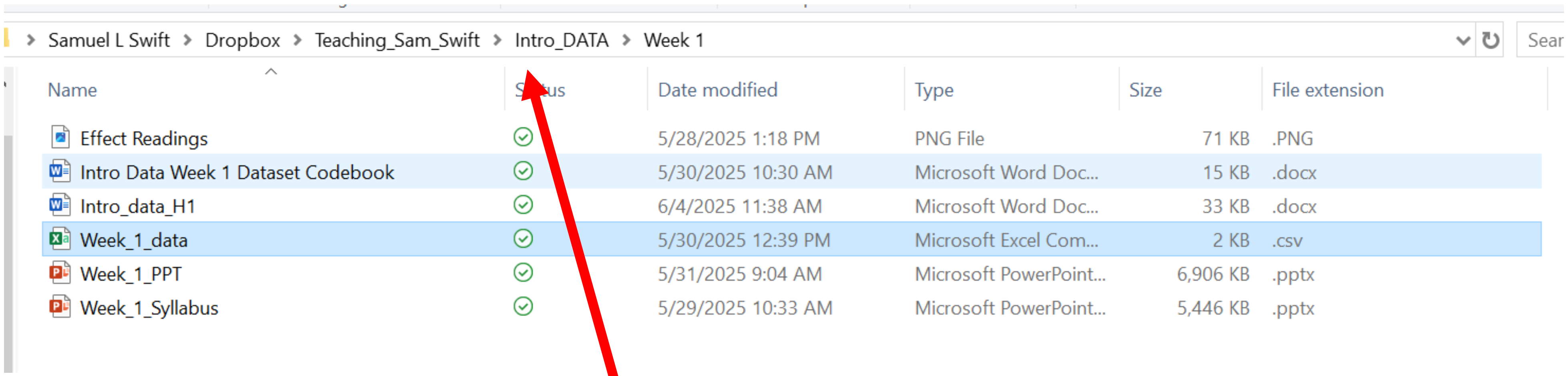


There are different ways to save the downloaded folder, which I will show on the video



WHERE YOU SAVE IT IS THE FILEPATH

Looks like this in windows file explorer



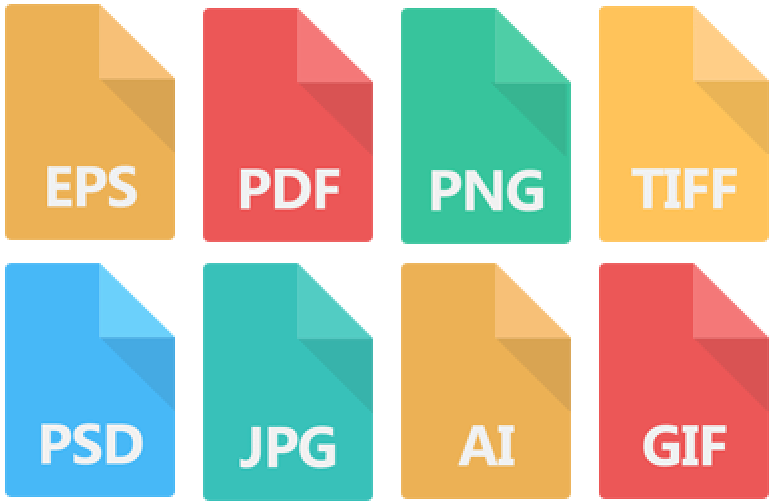
Looks like this in R studio

```
# Lets learn how to read in data
# TO MAKE THE LINE BELOW WORK YOU MUST CHANGE THE PATH- KEY CONCEPT FOR THE WEEK
# The path is where the data are, slashes are subfolders

data_week_1<-read.csv("C:/Users/samswift/Dropbox/Teaching_Sam_Swift/Intro_DATA/Week 1/Week_1_data.csv")
```

File extensions

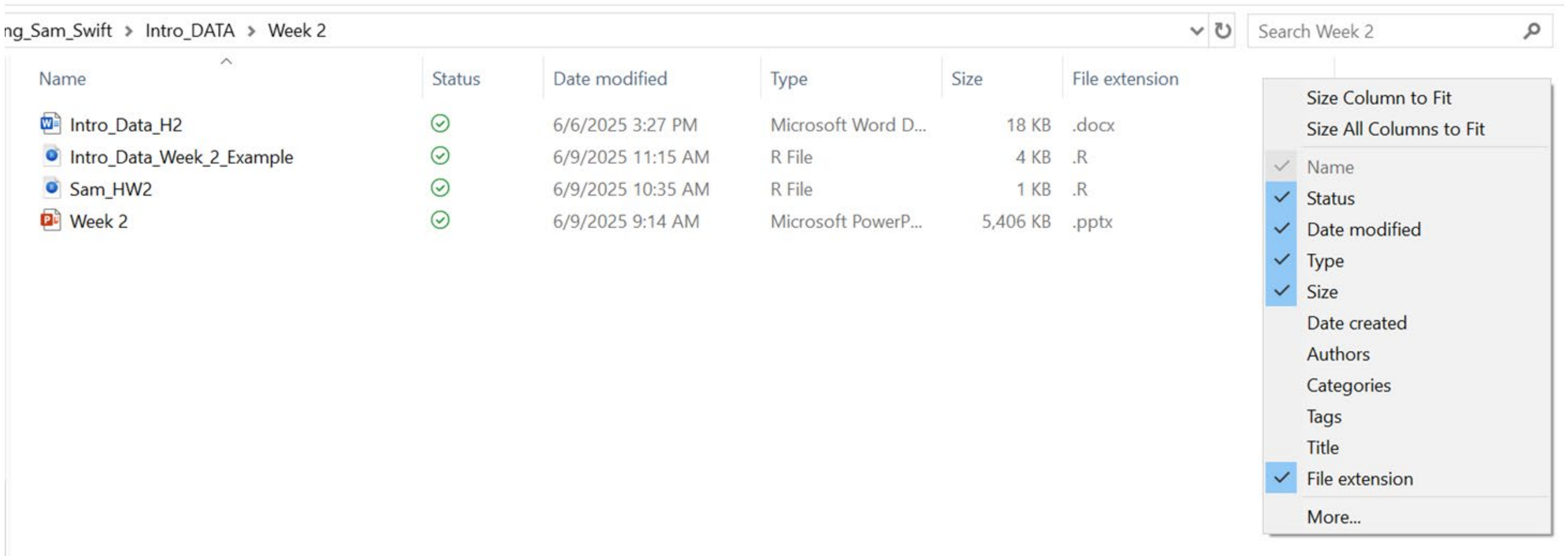
- In general, they tell us a lot of stuff about what type of file we are using
- In datasets, they can tell us about the statistical software we are using,
- The datasets file extensions also tell us the delimiter
- R packages can read in data from any statistical software



Add extensions to windows explorer/ finder

Windows explorer: https://lazyadmin.nl/win-11/show-file-extension-windows-11/

Finder (Mac): https://support.apple.com/guide/mac-help/show-or-hide-filename-extensions-on-mac-mchlp2304/mac



Anyway, Delimiters

Delimiters: Delimiters are characters that mark the boundaries between individual pieces of data within a file.

In this class we use CSV because the data are simple and do not cause problems for learners

.csv: Indicates comma-separated values.

.txt: Indicates a general text file, which can also be a delimited file.

.tsv: Indicates tab-separated values.

.psv: Indicates pipe-separated values.

CSV behind the scenes

What looks like this in Excel

	A	B	C	D	E	F	G
1	age	wildfiresm	heatdays	asthma	sbp	hea	
2	40	No	46	Yes	121.4	No	
3	60	No	26	No	127.3	No	No
4	23	No	2	No	117	No	No
5	76	Yes	53	No	144.8	Yes	Yes
6	51	No	0	No	109.5	No	No
7	59	No	23	No	111.2	Yes	No
8	52	No	0	Yes	122.2	Yes	No
9	23	Yes	15	Yes	107	No	Yes
10	34	No	60	No	143.9	No	No
11	78	No	60	No	119.1	No	No
12	66	No	23	No	108.7	No	No
13	18	No	11	Yes	116.6	Yes	No

```
"age","wildfiresmoke","heatdays","asthma","sbp","headaches","anxiety"
40,"No",46,"Yes",121.4,"No","No"
60,"No",26,"No",127.3,"No","No"
23,"No",2,"No",117,"No","No"
76,"Yes",53,"No",144.8,"Yes","Yes"
51,"No",0,"No",109.5,"No","No"
59,"No",23,"No",111.2,"Yes","No"
52,"No",0,"Yes",122.2,"Yes","No"
23,"Yes",15,"Yes",107,"No","Yes"
```

I will only use CSV in this class, and this is why

Looks like this in its raw form, which is easy for different software to understand

Note the commas!

Comments

- Comments start with this symbol #
  - (AKA octothorpe, pound sign, hashtag)
- These are used to document your code
- If you highlight these and run them nothing happens
- These are extremely useful for
  - Reproducible science
  - Trying to be a good teacher teaching people to code for first time
- You will type your HW answers as comments this week

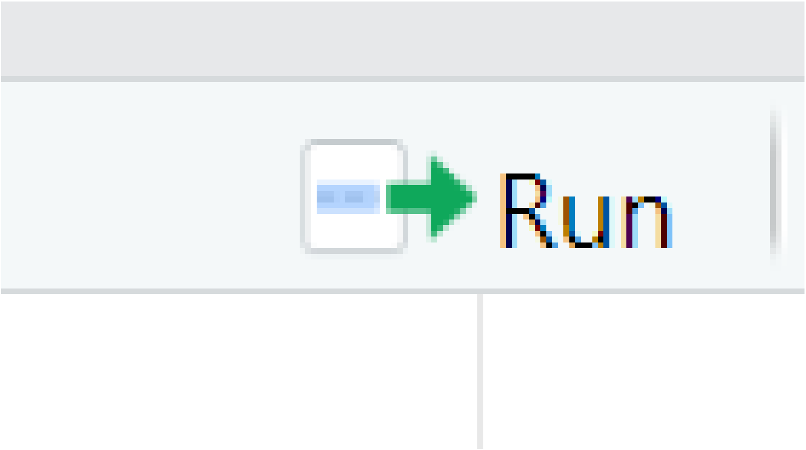
```
# here is the library function
library(tidyverse)

# Lets learn how to read in data
# TO MAKE THE LINE BELOW WORK YOU MUST CHANGE THE PATH- K
# The path is where the data are, slashes are subfolders

data_week_1<-read.csv("C:/Users/samswift/Dropbox/Teachinc
```

Running Code

- To run code, you can highlight a line of code and push the run button
- If there are open parenthesis on two lines, you need to highlight both
- You can also click on a line and push control+enter
- Command+enter on Mac



They made this button way too small!!!

Open Source: A Blessing and A Curse

- It is a scientific innovation that anyone, anywhere can write packages
- This increases reproducibility, shared knowledge, and innovation
- It also means sometimes syntax may vary slightly
- In extreme cases, packages can be wrong (none that we will be working with)

Questions?? Lets go to R